

LAMAP MALEKOULA ISLAND, Vanuatu

WMO: 915550

Lat: 16.422S

Lon: 167.809E

Elev: 26

StdP: 101.01

Time Zone: 11.00 (E11)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.9	20.8	14.3	10.2	22.7	15.5	11.0	22.8	10.5	24.4	9.6	24.9	1.8	120	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.9	31.1	27.0	30.7	26.8	30.3	26.5	27.9	30.4	27.5	30.0	27.2	29.7	3.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	23.0	29.9	26.7	22.4	29.4	26.4	21.9	29.1	89.4	30.5	87.5	30.2	86.1	29.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.1	8.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	27.4	27.5	27.4	26.9	26.1	25.3	24.8	24.5	25.2	25.9	26.5	27.1
	DBStd	1.51	0.91	0.96	0.86	0.87	0.94	1.14	1.30	1.47	1.34	1.18	1.03	0.96
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5917	541	491	540	507	499	458	458	449	457	492	496	529
	CDD18.3	2876	282	257	282	257	241	208	199	191	207	234	246	271
	CDH23.3	24719	2973	2733	2912	2436	1981	1386	1129	1021	1370	1859	2212	2709
CDH26.7	4930	836	792	786	491	242	108	73	79	144	269	417	693	
Wind	WSAvg	3.6	2.9	2.9	2.8	3.4	3.9	3.8	4.1	4.1	4.2	4.1	3.7	3.2
Precipitation	PrecAvg	1924	235	215	276	215	183	114	104	92	107	122	162	138
	PrecMax	3175	488	725	611	489	457	413	338	419	285	354	402	335
	PrecMin	1237	27	20	112	111	9	18	30	11	6	5	25	35
	PrecStd	480	120	140	128	78	118	83	66	87	82	98	102	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.5	31.7	31.4	30.7	29.8	29.2	29.2	29.3	29.7	30.1	30.9	31.2
		MCWB	27.1	27.6	27.1	27.2	26.3	25.8	25.5	25.0	25.6	25.8	26.8	26.4
	2%	DB	30.9	31.0	30.8	30.2	29.2	28.4	28.2	28.2	29.0	29.4	30.1	30.6
		MCWB	26.9	27.1	27.1	26.8	25.9	25.1	24.6	24.4	25.1	25.3	26.1	26.2
	5%	DB	30.4	30.5	30.3	29.6	28.7	27.9	27.4	27.5	28.2	28.7	29.5	30.1
		MCWB	26.6	26.7	26.8	26.4	25.5	24.7	24.0	23.7	24.5	24.8	25.7	26.1
	10%	DB	29.9	30.0	29.8	29.0	28.0	27.3	26.7	26.7	27.5	28.2	28.9	29.6
		MCWB	26.2	26.4	26.5	26.0	25.0	24.3	23.5	23.1	24.0	24.5	25.3	25.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	28.4	28.1	27.9	27.3	26.6	26.2	25.7	26.4	26.7	27.6	27.9
		MCDB	30.7	31.0	30.6	30.3	29.2	28.3	28.3	28.1	29.1	29.2	30.2	30.4
	2%	WB	27.7	27.7	27.6	27.2	26.5	25.9	25.4	25.1	25.7	26.1	26.8	27.3
		MCDB	30.1	30.4	30.1	29.6	28.5	27.5	27.3	27.4	28.3	28.6	29.3	29.9
	5%	WB	27.3	27.2	27.2	26.8	26.0	25.4	24.8	24.6	25.2	25.6	26.3	26.8
		MCDB	29.6	29.9	29.7	29.1	28.1	27.1	26.6	26.8	27.5	28.0	28.7	29.5
	10%	WB	26.9	26.8	26.8	26.4	25.6	24.9	24.3	24.0	24.7	25.1	25.8	26.3
		MCDB	29.2	29.4	29.2	28.5	27.5	26.8	26.2	26.2	26.9	27.4	28.2	28.9
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.9	4.8	4.3	3.7	3.9	3.9	4.3	4.3	4.4	4.5	4.8
		MCDBR	5.5	5.5	5.4	5.0	4.6	4.5	4.6	5.1	5.1	5.0	5.4	5.6
		MCWBR	2.8	2.8	2.9	2.7	2.7	2.5	2.7	2.8	2.8	2.8	2.9	2.9
	5% WB	MCDBR	4.7	5.2	5.0	4.6	4.4	3.9	3.9	4.5	4.6	4.4	5.0	5.1
		MCWBR	2.9	2.9	2.8	2.7	2.7	2.4	2.6	2.8	2.7	2.7	2.9	2.9
Clear-Sky Solar Irradiance	taub	0.408	0.400	0.393	0.387	0.394	0.411	0.399	0.393	0.405	0.406	0.410	0.400	
	taud	2.533	2.565	2.592	2.610	2.559	2.492	2.516	2.530	2.484	2.502	2.505	2.540	
	Ebn at Noon	937	936	921	892	849	813	836	873	896	919	929	944	
	Edh at Noon	112	107	101	94	93	96	96	100	111	113	114	111	
All-Sky Solar Radiation	RadAvg	6.12	5.91	5.47	4.68	4.01	3.55	3.73	4.35	5.18	5.62	5.96	6.36	
	RadStd	0.58	0.48	0.46	0.42	0.40	0.28	0.26	0.31	0.47	0.53	0.55	0.43	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.33	N/A	N/A	N/A	N/A	

Nomenclature: See separate page